

Customer Behavior Analysis

Internship Project

By: Rudrashetty Nandini



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Data Analytics Internship Project



Submitted By

Rudrashetty Nandini



Project Objective

The objective of this project is to analyze customer purchasing behavior, spending patterns, and engagement trends using data analytics and visualization techniques.



Tools & Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Google Colab



Key Analysis Performed

- Data Cleaning & Preprocessing
- Customer Spending Analysis
- Age Group Analysis
- Purchase Trend Analysis
- Customer Segmentation
- Visualization & Insights



Expected Outcome

The analysis helps identify customer preferences, spending behavior, and business opportunities for improving customer engagement and sales performance.

Customer Behavior Analysis

```
from google.colab import files
```

```
uploaded = files.upload()
```

No file chosen Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.
Saving ecommerce_customer_data_custom_ratios.csv to ecommerce_customer_data_c

Import Libraries

```
import os
```

```
os.listdir('/content')
```

```
['.config',  
'ecommerce_customer_data_large.csv',  
'ecommerce_customer_data_custom_ratios (1).csv',  
'ipynb_checkpoints',  
'ecommerce_customer_data_custom_ratios.csv',  
'ecommerce_customer_data_large (1).csv',  
'sample_data']
```

Load Dataset

```
import pandas as pd
```

```
df1 = pd.read_csv('/content/ecommerce_customer_data_large.csv')
```

```
df2 = pd.read_csv('/content/ecommerce_customer_data_custom_ratios.csv')
```

```
print(df1.columns)
```

```
print(df2.columns)
```

```
Index(['Customer ID', 'Purchase Date', 'Product Category', 'Product Price',  
      'Quantity', 'Total Purchase Amount', 'Payment Method', 'Customer Age',  
      'Returns', 'Customer Name', 'Age', 'Gender', 'Churn'],  
      dtype='object')  
Index(['Customer ID', 'Purchase Date', 'Product Category', 'Product Price',  
      'Quantity', 'Total Purchase Amount', 'Payment Method', 'Customer Age',  
      'Returns', 'Customer Name', 'Age', 'Gender', 'Churn'],  
      dtype='object')
```

```
print(df1.head())
```

```
print(df2.head())
```

	Customer ID	Purchase Date	Product Category	Product Price	Quantity
0	44605	2023-05-03 21:30:02	Home	177	1
1	44605	2021-05-16 13:57:44	Electronics	174	3
2	44605	2020-07-13 06:16:57	Books	413	1
3	44605	2023-01-17 13:14:36	Electronics	396	3
4	44605	2021-05-01 11:29:27	Books	259	4

	Total Purchase Amount	Payment Method	Customer Age	Returns	Customer Name
0	2427	PayPal	31	1.0	John Rivera
1	2448	PayPal	31	1.0	John Rivera
2	2345	Credit Card	31	1.0	John Rivera
3	937	Cash	31	0.0	John Rivera
4	2598	PayPal	31	1.0	John Rivera

	Age	Gender	Churn
0	31	Female	0
1	31	Female	0
2	31	Female	0
3	31	Female	0
4	31	Female	0

	Customer ID	Purchase Date	Product Category	Product Price	Quantity
0	46251	2020-09-08 09:38:32	Electronics	12	3
1	46251	2022-03-05 12:56:35	Home	468	4
2	46251	2022-05-23 18:18:01	Home	288	2
3	46251	2020-11-12 13:13:29	Clothing	196	1
4	13593	2020-11-27 17:55:11	Home	449	1

	Total Purchase Amount	Payment Method	Customer Age	Returns	\
0	740	Credit Card	37	0.0	
1	2739	PayPal	37	0.0	
2	3196	PayPal	37	0.0	
3	3509	PayPal	37	0.0	
4	3452	Credit Card	49	0.0	

	Customer Name	Age	Gender	Churn
0	Christine Hernandez	37	Male	0
1	Christine Hernandez	37	Male	0
2	Christine Hernandez	37	Male	0
3	Christine Hernandez	37	Male	0
4	James Grant	49	Female	1

```
print(df1.columns)
```

```
print(df2.columns)
```

```
Index(['Customer ID', 'Purchase Date', 'Product Category', 'Product Price',
      'Quantity', 'Total Purchase Amount', 'Payment Method', 'Customer Age',
      'Returns', 'Customer Name', 'Age', 'Gender', 'Churn'],
      dtype='object')
Index(['Customer ID', 'Purchase Date', 'Product Category', 'Product Price',
      'Quantity', 'Total Purchase Amount', 'Payment Method', 'Customer Age',
      'Returns', 'Customer Name', 'Age', 'Gender', 'Churn'],
      dtype='object')
```

```
import pandas as pd

df1 = pd.read_csv('/content/ecommerce_customer_data_large.csv')

df2 = pd.read_csv('/content/ecommerce_customer_data_custom_ratios.csv')

df = pd.concat([df1, df2], ignore_index=True)

print(df.shape)
```

```
(500000, 13)
```

```
df.head()
```

	Customer ID	Purchase Date	Product Category	Product Price	Quantity	Total Purchase Amount	Payment Method	Customer Age
0	44605	2023-05-03 21:30:02	Home	177	1	2427	PayPal	
1	44605	2021-05-16 13:57:44	Electronics	174	3	2448	PayPal	
2	44605	2020-07-13 06:16:57	Books	413	1	2345	Credit Card	
3	44605	2023-01-17	Electronics	396	3	937	Cash	

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 500000 entries, 0 to 499999
Data columns (total 13 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Customer ID                          500000 non-null  int64
1   Purchase Date                        500000 non-null  object
2   Product Category                     500000 non-null  object
3   Product Price                        500000 non-null  int64
4   Quantity                             500000 non-null  int64
5   Total Purchase Amount                500000 non-null  int64
6   Payment Method                      500000 non-null  object
7   Customer Age                         500000 non-null  int64
8   Returns                             405022 non-null  float64
9   Customer Name                       500000 non-null  object
10  Age                                  500000 non-null  int64
11  Gender                              500000 non-null  object
12  Churn                               500000 non-null  int64
dtypes: float64(1), int64(7), object(5)
memory usage: 49.6+ MB
```



Data Cleaning

```
df.isnull().sum()
```

	0
Customer ID	0
Purchase Date	0
Product Category	0
Product Price	0
Quantity	0
Total Purchase Amount	0
Payment Method	0
Customer Age	0
Returns	94978
Customer Name	0
Age	0
Gender	0
Churn	0

dtype: int64

```
df.drop_duplicates(inplace=True)
```

```
df.dropna(inplace=True)
```

```
df.columns = df.columns.str.strip()
```

```
df['Total Purchase Amount']
```

Total Purchase Amount	
0	2427
1	2448
2	2345
3	937
4	2598
...	...
499995	2187
499996	3615
499997	2466
499998	3668
499999	2370

405022 rows × 1 columns

dtype: int64

```
df['Purchase Date'] = pd.to_datetime(df['Purchase Date'])
```

```
df['Month'] = df['Purchase Date'].dt.month
```

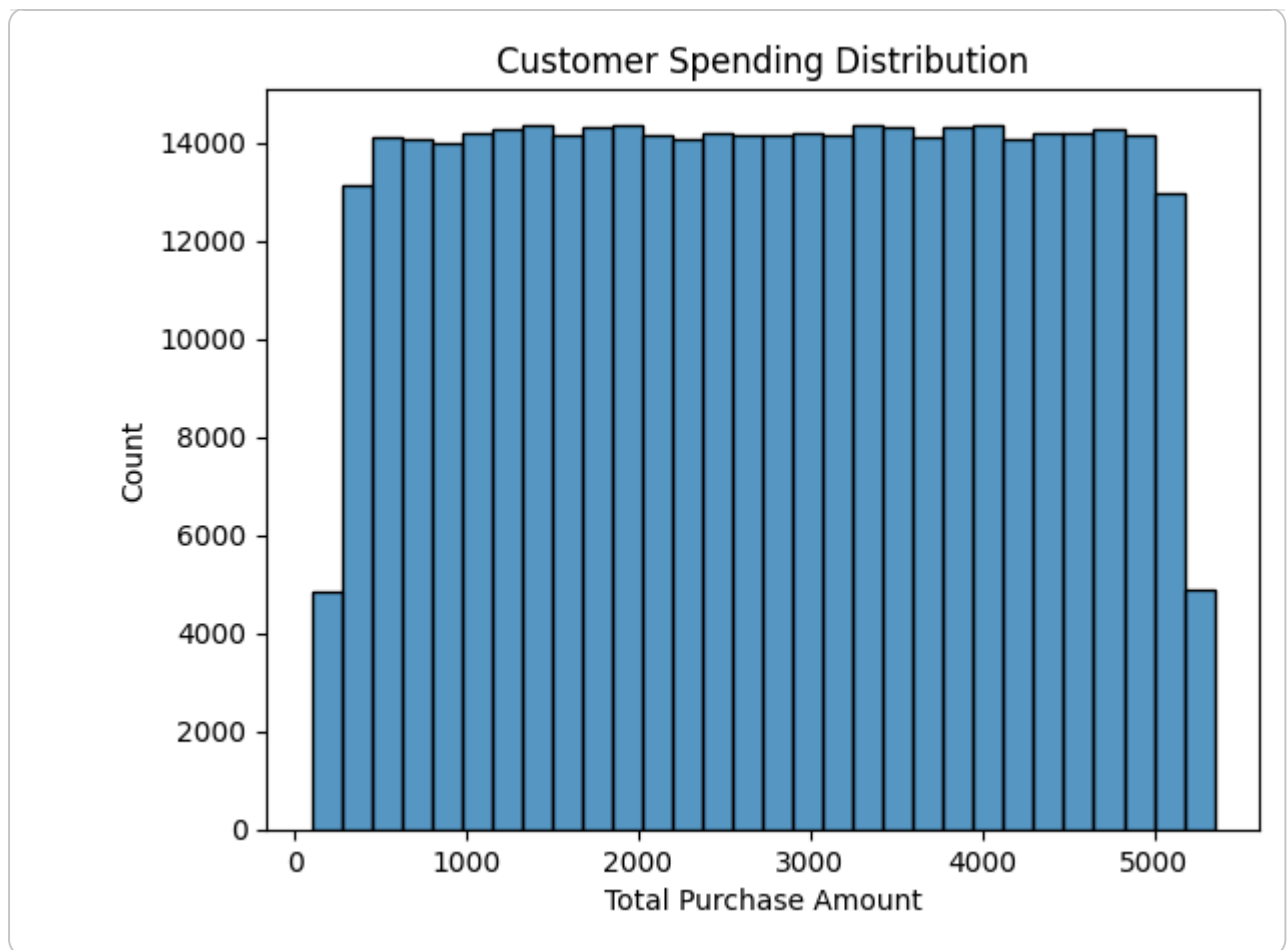
✓ Exploratory Data Analysis

```
import matplotlib.pyplot as plt
import seaborn as sns

sns.histplot(df['Total Purchase Amount'], bins=30)

plt.title('Customer Spending Distribution')

plt.show()
```



Spending Distribution Graph Insights

Insights

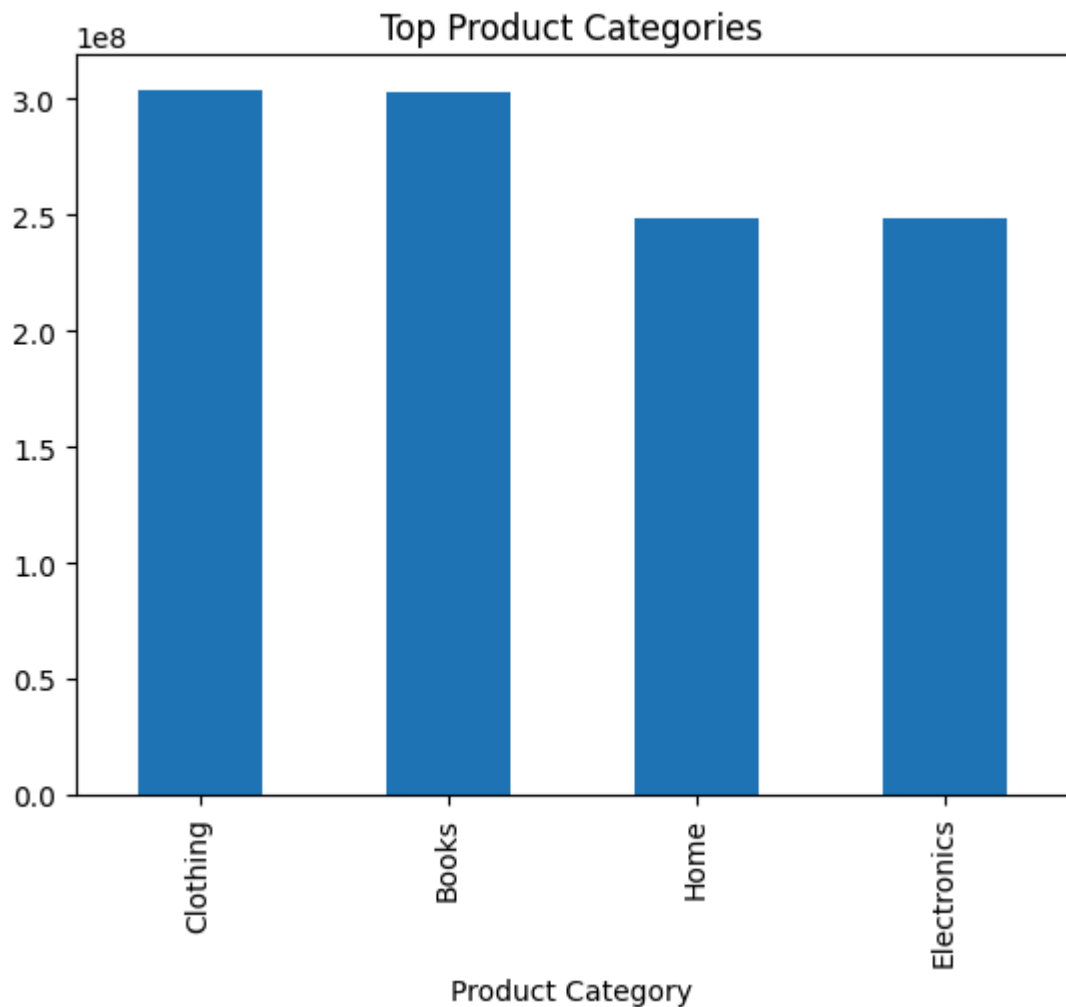
- Most customers spend moderate amounts on purchases.
- A smaller group of customers contributes significantly higher revenue.
- The spending distribution is positively skewed, indicating the presence of high-value customers.
- These high-spending customers can be targeted with premium offers and loyalty programs.

```
top_products = df.groupby('Product Category')['Total Purchase Amount'].sum()

top_products.plot(kind='bar')

plt.title('Top Product Categories')

plt.show()
```



Top Product Categories Graph Insights

Insights

- Certain product categories generate significantly higher revenue than others.
- High-performing categories should receive more marketing focus and inventory support.
- Low-performing categories may require promotional campaigns or pricing adjustments.
- Understanding product demand helps improve business strategy and sales optimization.

```
gender_sales = df.groupby('Gender')['Total Purchase Amount'].sum()

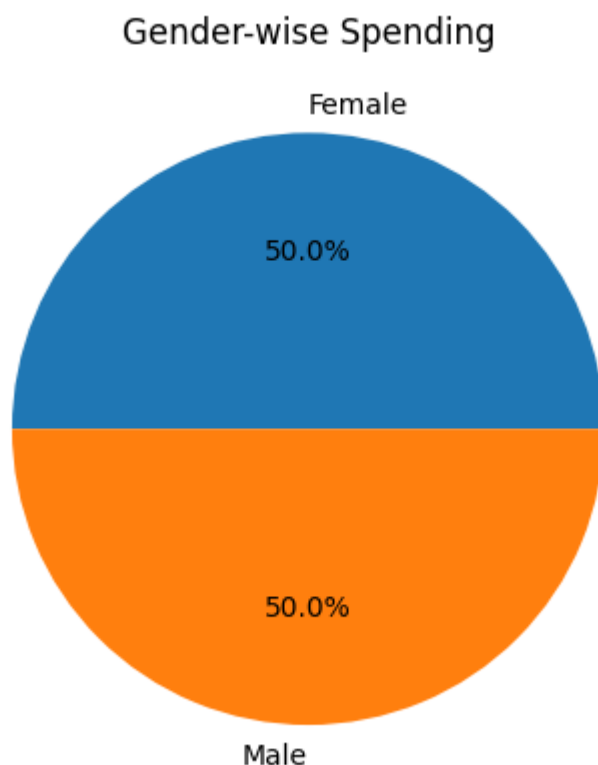
gender_sales.plot(kind='pie', autopct='%1.1f%%')

plt.ylabel('')

plt.title('Gender-wise Spending')
```



```
plt.show()
```



Gender-wise Spending Insights

Insights

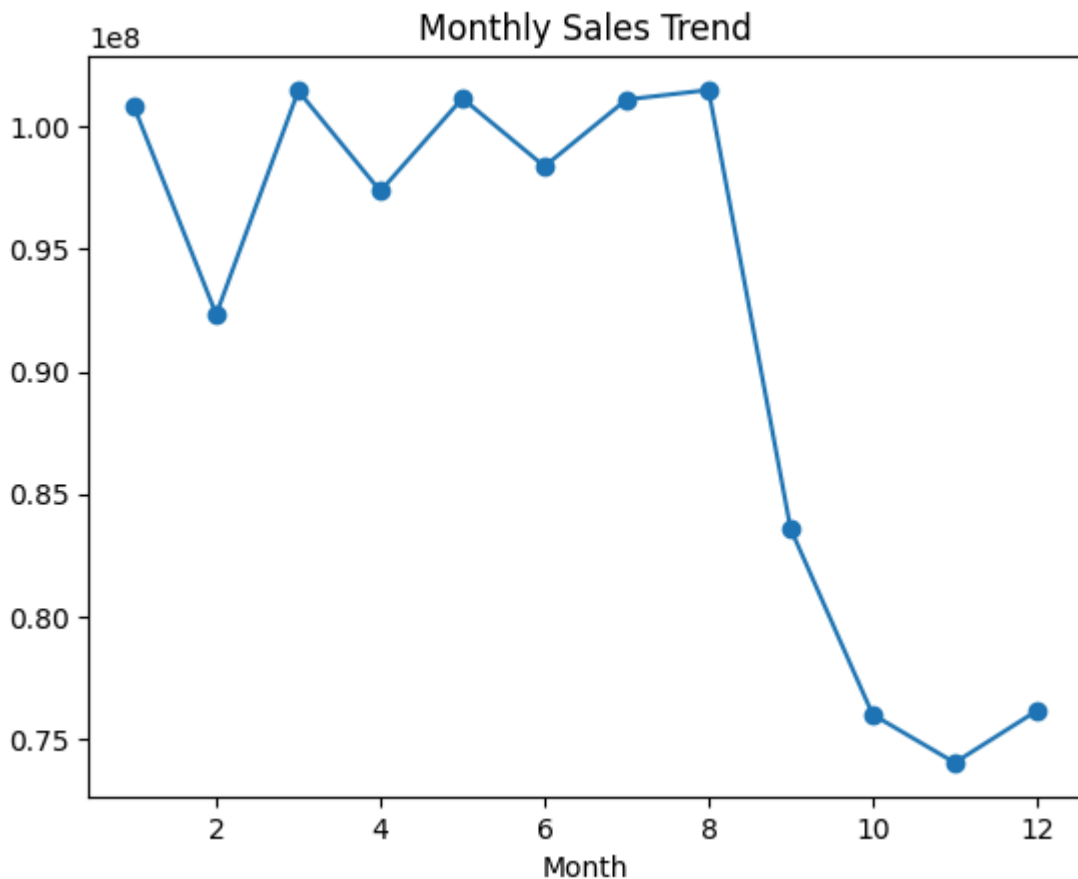
- Spending behavior varies across gender groups.
- One gender group contributes a larger share of total revenue.
- Personalized marketing campaigns can improve customer engagement.
- Gender-based customer preferences can help optimize product recommendations.

```
monthly_sales = df.groupby('Month')['Total Purchase Amount'].sum()

monthly_sales.plot(marker='o')

plt.title('Monthly Sales Trend')

plt.show()
```



Monthly Sales Trend Insights

Insights

- Sales fluctuate across different months, indicating seasonal purchasing behavior.
- Certain months show peak customer activity and higher revenue generation.
- Businesses can increase marketing efforts during high-sales periods.
- Lower-sales months may require promotional campaigns to improve engagement.

```
from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler
```

```
features = df[['Total Purchase Amount', 'Quantity']]
```

```
scaler = StandardScaler()

scaled_features = scaler.fit_transform(features)
```

```
kmeans = KMeans(n_clusters=3, random_state=42)
```

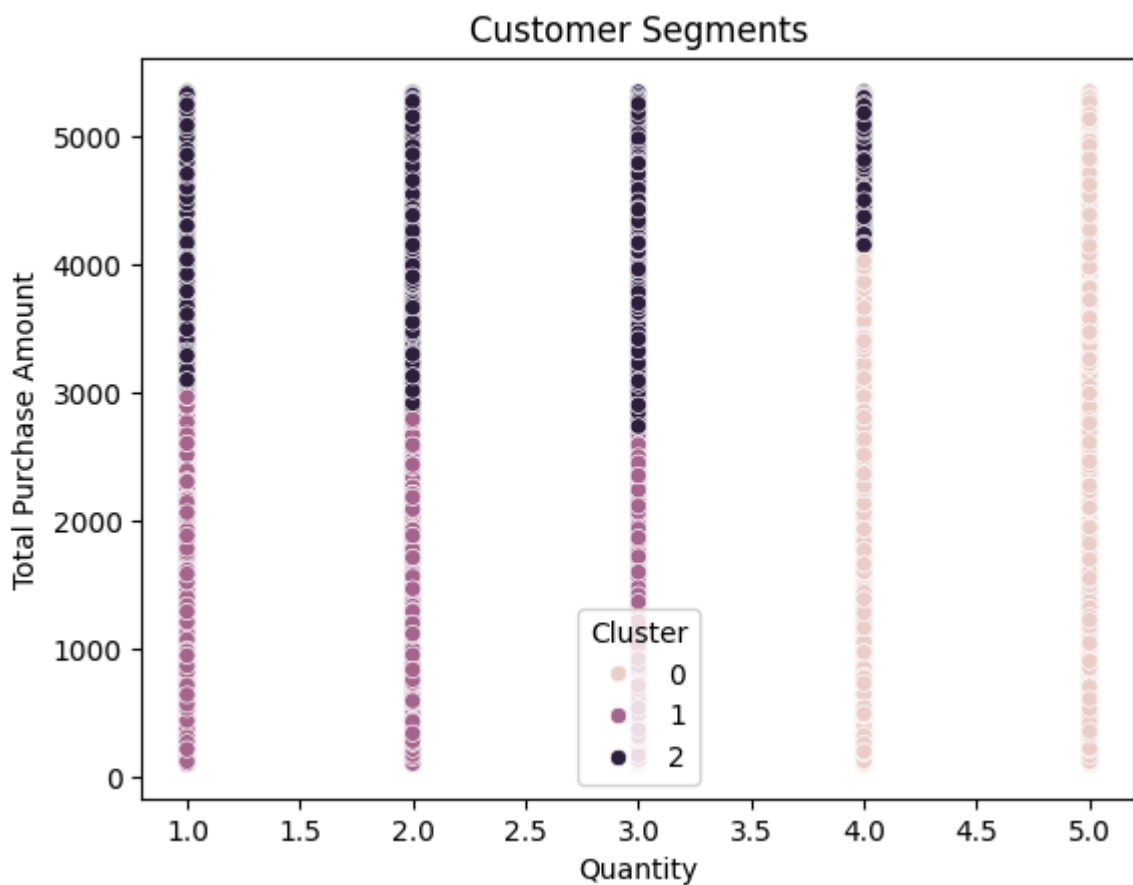
```
df['Cluster'] = kmeans.fit_predict(scaled_features)
```

```
sns.scatterplot(  
    x=df['Quantity'],  
    y=df['Total Purchase Amount'],  
    hue=df['Cluster']  
)
```

```
plt.title('Customer Segments')
```

```
plt.show()
```

```
/usr/local/lib/python3.12/dist-packages/IPython/core/pylabtools.py:151: UserWarning  
fig.canvas.print_figure(bytes_io, **kw)
```



Customer Segmentation Insights

Insights

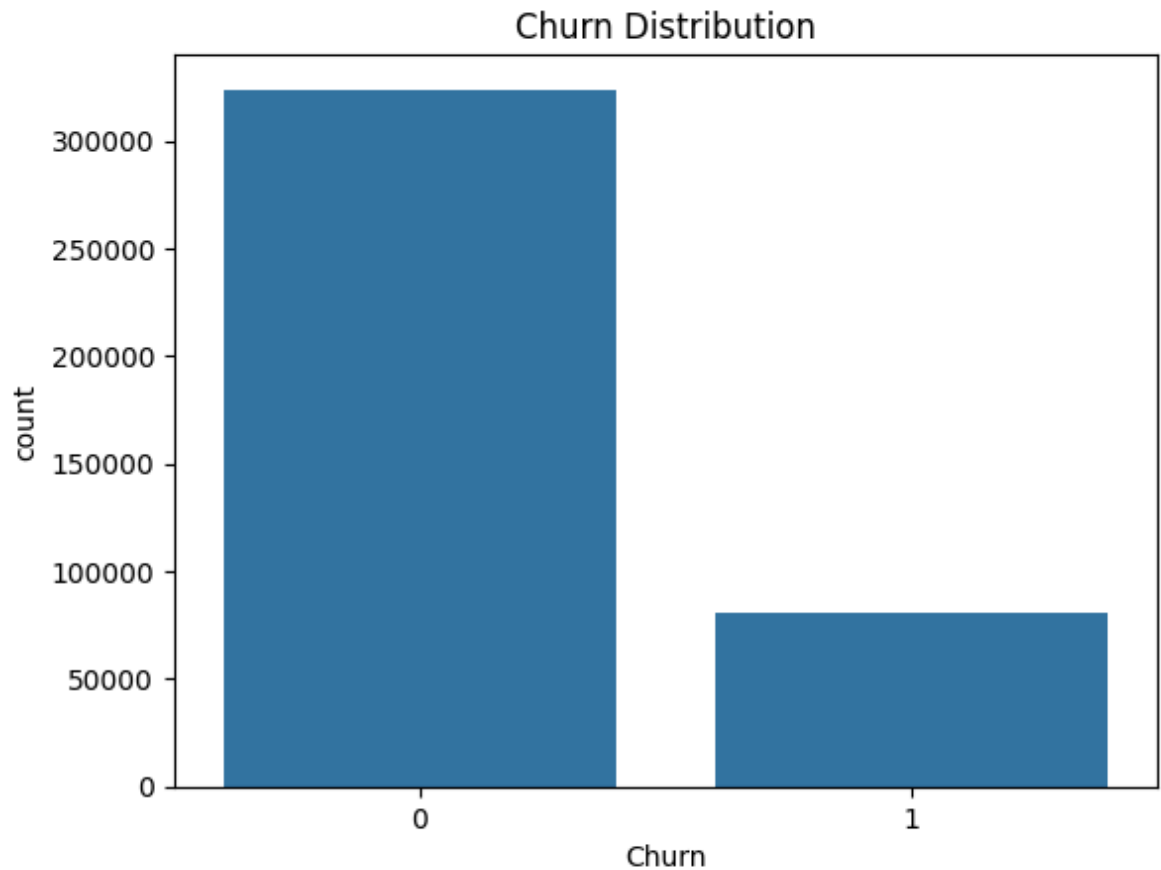
- Customers were successfully grouped into different spending segments.
- One cluster represents high-value customers with larger purchase amounts.
- Another cluster contains low-spending customers with lower engagement.
- Segment-specific marketing strategies can improve customer retention and revenue.

- High-value customers should be prioritized through loyalty rewards and personalized services.

```
sns.countplot(x='Churn', data=df)

plt.title('Churn Distribution')

plt.show()
```



Churn Distribution Insights

Insights

- A noticeable percentage of customers are marked as churned.
- Churned customers indicate potential customer retention issues.
- Businesses should focus on re-engagement strategies to reduce churn.
- Personalized discounts and loyalty programs may help retain at-risk customers.
- Reducing churn can significantly improve long-term profitability.

```
churn_percent = df['Churn'].value_counts(normalize=True) * 100

print(churn_percent)
```

```
Churn
0    79.974421
1    20.025579
Name: proportion, dtype: float64
```

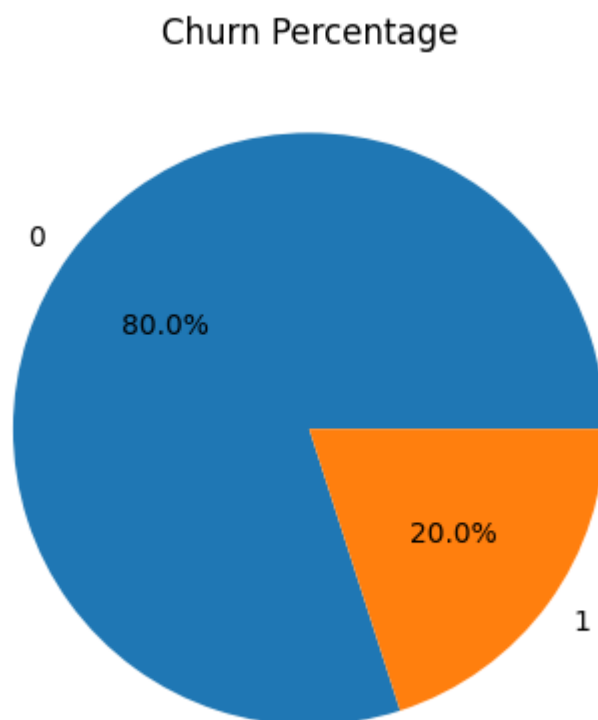
```
import matplotlib.pyplot as plt

churn_percent.plot(kind='pie', autopct='%1.1f%%')

plt.ylabel('')

plt.title('Churn Percentage')

plt.show()
```



Churn Analysis Insights

A significant percentage of customers are marked as churned. This indicates potential customer retention issues.

Customers with churn risk should be targeted with:

personalized offers loyalty rewards re-engagement campaigns Reducing churn can improve long-term revenue and customer satisfaction.

Key Findings

- High-value customer groups were identified using clustering techniques.
- Customer purchasing behavior varies across different segments.
- Seasonal trends influence overall sales performance.
- Churn analysis revealed customers at risk of disengagement.
- Product category analysis identified high-performing products.
- Data-driven marketing strategies can improve customer retention and revenue growth.

Recommendations

1. Introduce loyalty programs for repeat and high-value customers.
2. Provide personalized product recommendations based on purchase history.